

August 4, 2026

Dear Hiring Team,

I am writing to apply for the Research Engineer position with Cohere Labs. With a Master of Science in Computer Science from the University of Calgary and applied experience in Python, PyTorch, scalable research software, model evaluation, and data processing, I am excited by the opportunity to help turn machine learning research ideas into working systems.

My graduate research focused on building machine learning and data-management systems for large geospatial and Earth observation datasets. In my digital elevation model super-resolution project, I designed Python and PyTorch workflows for data collection, preprocessing, model training, and evaluation. I trained deep ResNet models for 4x terrain reconstruction and compared model quality using RMSE, slope, TRI, TPI, and wavelet-based metrics. This work required careful experimentation, debugging, analysis of model behavior, and attention to whether results generalized beyond convenient random splits.

I also developed C++ and Python software for scalable processing, storage, and retrieval of satellite imagery through my research work with BigGeo, Mitacs, and the University of Calgary. I built DGGS-based encoding pipelines, designed tensor-based storage for multiresolution retrieval, and improved processing performance with C++ multithreading. These projects strengthened the engineering habits I would bring to Cohere Labs: writing modular code, testing ideas through working prototypes, documenting technical decisions, and making research outputs usable by others.

More recently, I have been building an agentic job-search automation platform using OpenClaw, n8n, LLM workflows, Docker, SearXNG, Google Sheets, Google Drive, and Telegram integrations. The project has given me hands-on exposure to orchestrating LLM-based workflows, connecting models with search and external tools, and designing automation that needs to remain practical and auditable. While I am still growing in areas such as large-scale LLM finetuning, RLHF, and distributed training, my background gives me a strong base for learning quickly in a fast-moving research environment.

Cohere Labs stands out to me because the role values curiosity, strong fundamentals, clean implementation, ablations, evaluation, and close collaboration between researchers and engineers. I would be glad to contribute my PyTorch, Python, research software, data-processing, and technical communication experience while continuing to grow into deeper NLP and frontier AI systems work.

Thank you for considering my application. I would welcome the opportunity to discuss how my research engineering experience can contribute to Cohere Labs.

Sincerely,
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